

Power BI Report Documentation

A/B Testing — Landing Page Optimization

Final report documentation for the four-page Power BI dashboard, covering report architecture, page purpose, measures, interactions, tooltips, performance, and QA.

1. Project Overview

The report evaluates an A/B experiment comparing a Control landing-page experience with a Treatment experience. The dashboard is designed to move from executive results to customer/segment behavior, statistical validation, and finally a business decision.

Item	Final configuration
Report type	Power BI interactive A/B testing dashboard
Experiment	Control vs Treatment landing-page test
Users	294,478 unique users
Control	146,926 users
Treatment	147,552 users
Conversions	43,928
Primary outcome	Conversion Rate
Final pages	4
Navigation	Implemented
Report-page tooltips	Device, Location, Segment
Reset/filter UX	Implemented

2. Final GitHub Project Structure

Recommended repository structure:

```
ab-testing-landing-page-optimization/  
├── README.md  
├── powerbi/  
│   ├── AB_Testing_Landing_Page_Optimization.pbix  
│   ├── Power_BI_Report_Documentation.pdf  
│   └── screenshots/  
│       ├── 01-executive-experiment.png  
│       ├── 02-customer-segment-intelligence.png  
│       ├── 03-experiment-statistical-analysis.png  
│       └── 04-decision-intelligence.png  
├── analysis/  
│   ├── ab_testing_analysis.ipynb  
│   └── statistical_results.csv  
├── documentation/  
│   ├── DAX_Measures.md  
│   ├── Data_Model.md  
│   └── Performance_Analysis.md  
├── data/  
│   └── README.md  
└── .gitignore
```

Do not commit raw or sensitive source data unless redistribution is permitted. If the dataset is too large or licensed for local use only, keep it out of GitHub and document how it can be obtained.

3. Report Architecture

Page	Name	Primary question
01	Executive Experiment	Did Treatment outperform Control?
02	Customer & Segment Intelligence	Who responded, how did they behave, and where did Treatment perform best?
03	Experiment & Statistical Analysis	Are the observed differences statistically reliable?
04	Decision Intelligence	What should the business do based on the evidence?

4. Page 1 — Executive Experiment

Purpose: provide an immediate executive-level summary of the experiment.

Key elements:

- Total Users / experiment scale
- Control and Treatment conversion performance
- Conversion lift
- Revenue / visitor or revenue outcome
- Primary statistical significance indicator
- High-level comparison visuals

Headline result used in the report: Treatment conversion is 17.95% versus 11.87% for Control, a 51.18% relative lift and a 6.08 percentage-point absolute increase.

5. Page 2 — Customer & Segment Intelligence

Purpose: understand behavioral and segment-level differences without duplicating the executive page.

Key analytical areas:

- Device-level Control vs Treatment conversion
- Location performance
- Location × Device heatmap
- Age / purchase behavior relationship
- Segment and audience distribution

This page deliberately uses multiple visual types rather than relying only on clustered columns or bar charts.

6. Page 3 — Experiment & Statistical Analysis

Purpose: establish whether the observed business differences are statistically meaningful.

Metric	Analysis / test	Reported interpretation
Conversion	Two-proportion Z-test	Statistically significant
Revenue	Welch's t-test	Statistically significant
Session Duration	Welch's t-test	Not statistically significant
Pages Visited	Welch's t-test	Not statistically significant

The report uses the statistical analysis to distinguish genuine experiment effects from differences that may be attributable to sampling variation.

7. Page 4 — Decision Intelligence

Purpose: convert the experiment evidence into an actionable business conclusion.

Core elements:

- Experiment Winner KPI
- Treatment Conversion Rate
- Conversion Lift
- P-value
- Revenue significance result
- Experiment Verdict
- Key Findings
- Business Recommendation
- Interactive Decomposition Tree

Decision framing: Treatment outperforms Control on the primary conversion outcome, with statistical evidence supporting the observed conversion improvement. Revenue also shows a statistically significant difference, while session duration and pages visited do not.

8. Core DAX Measures

Representative measures used throughout the report:

Total Users

Total Users = DISTINCTCOUNT(FactABTest[User ID])

Average Applicants

Average Applicants = AVERAGE(FactABTest[Applicant Count])

Conversion Rate

Conversion Rate = DIVIDE([Converted Users], [Total Users])

Conversion Lift

Conversion Lift % = DIVIDE([Treatment Conversion Rate] - [Control Conversion Rate], [Control Conversion Rate])

Conversion Lift (pp)

Conversion Lift (pp) = [Treatment Conversion Rate] - [Control Conversion Rate]

Experiment Winner

Experiment Winner = IF([Treatment Conversion Rate] > [Control Conversion Rate], "Treatment", "Control")

Use the actual PBIX measure definitions as the authoritative source if any formula above differs from the final model. This documentation intentionally records the final analytical intent rather than inventing model fields that were not confirmed.

9. Tooltip Architecture

Tooltip	Used for	Dynamic context
TT – Device	Device visuals	Hovered Device Type
TT – Location	Location visuals	Hovered Location
TT – Segment	Segment / scatter analysis	Hovered segment context such as Age and Group

All three report-page tooltips were tested and confirmed to respect the hovered visual's filter context. The tooltip system should not be expanded unless a future visual genuinely benefits from one.

10. Interaction & UX

- Four-page navigation is implemented.
- Reset/filter UX is implemented.
- Cross-filtering and cross-highlighting were reviewed.
- Decomposition Tree is used for interactive driver exploration.
- Report-page tooltips provide contextual detail without adding permanent visual clutter.

11. Performance Notes

Performance Analyzer identified Engagement vs Purchase Value as the main performance hotspot. Its recorded visual lifecycle was approximately 783 ms, with approximately 609 ms of query time and a query returning up to roughly 3,501 sampled points. The Age vs Purchase Behavior visual was not the main bottleneck; its query grouped by Group and Age and returned a much smaller result set.

Because the project is being wrapped up, the recommended approach is to document this as a known optimization opportunity rather than redesign the finished report. If performance becomes a priority later, aggregate the engagement/purchase scatter to a smaller analytical grain and retest with Performance Analyzer.

12. QA Checklist

- ☐ All four report pages load successfully.
- ☐ Navigation works in both directions.
- ☐ Reset filters returns pages to the intended default state.
- ☐ Device tooltip changes with the hovered device.
- ☐ Location tooltip changes with the hovered location.
- ☐ Segment tooltip changes with the hovered segment.
- ☐ Control/Treatment labels are consistent across pages.
- ☐ Conversion Rate, lift, p-value, and significance labels are consistent.
- ☐ Decomposition Tree responds to the available dimensions.
- ☐ No visual displays unrelated values because of missing filter context.
- ☐ Final screenshots match the published report.

13. Portfolio / GitHub Presentation

Recommended README sections:

- Project Overview
- Business Problem
- Dataset & Experiment Design
- Key Questions
- Tools & Technologies
- Power BI Report Walkthrough
- Statistical Methodology
- Key Findings
- Business Recommendation
- Performance Considerations
- Repository Structure
- How to Reproduce

14. Final Project Status

Component	Status
A/B test analysis	Complete
Power BI data model	Complete
Four dashboard pages	Complete
Navigation	Complete
Tooltips	Complete
Bookmarks / reset UX	Complete
Performance analysis	Complete
Known performance hotspot documented	Complete
GitHub packaging	Ready

Final objective: publish a clean, reproducible portfolio project that demonstrates experimentation, statistical reasoning, Power BI modeling, DAX, interactive UX, and business decision-making.